

Amazing 🎉 — great job getting the **entire project running end-to-end**.

Below is a **clean, professional, future-proof README.md** you can directly copy-paste into your project root. It's written so that **April-you** (or your mentor/examiner) can run it smoothly in **one go** on VS Code.

👉 Multimodal Real-Time Sign Language to Text & Speech Translator

A **real-time multimodal AI system** that converts **American Sign Language (ASL)** gestures into **natural English text**, refines it using a **local LLM (LM Studio)**, optionally **translates** it to other languages, and **speaks it aloud** using **Text-to-Speech (TTS)**.

✨ Features

- 🎥 Real-time ASL letter recognition using **MediaPipe + ML**
 - 🧠 Intelligent sentence refinement using **local LLM (LM Studio)**
 - 🌐 Optional multi-language translation
 - 🗣️ Natural Text-to-Speech output (Piper TTS)
 - 💾 Session storage in **MongoDB**
 - 💻 Fully local, privacy-preserving system
-

🧱 Tech Stack

- **Frontend:** Streamlit
 - **Computer Vision:** OpenCV, MediaPipe
 - **ML Model:** Custom ASL Keypoint Classifier
 - **NLP / LLM:** LM Studio (OpenAI-compatible local API)
 - **TTS:** Piper TTS
 - **Database:** MongoDB
 - **Language:** Python 3.10+
-

📁 Project Structure

Multimodal-Realtime-Sign-Language-to-Text-and-Speech-Translator/
|

```
├── streamlit_app.py      # Main Streamlit app
├── nlp_refiner.py        # LLM-based text refinement
├── tts_manager.py        # Text-to-Speech handler
├── asl_classifier/       # ASL model & CSV
├── utils/                # FPS + helpers
├── voices/               # Piper TTS voice models
├── requirements.txt
├── .env
└── README.md
```

One-Time Setup (Only Once)

1 Install Python

- Install **Python 3.10 or later**
 - During install:  *Add Python to PATH*
-

2 Open Project in VS Code

File → Open Folder → Select project folder

3 Create & Activate Virtual Environment

```
python -m venv venv
.\venv\Scripts\Activate.ps1
```

You should see:

```
(venv)
```

4 Install Dependencies

```
pip install -r requirements.txt
```

5 Install & Setup MongoDB

- Install MongoDB Community Server
- Start MongoDB service

- Default URI used:

mongodb://127.0.0.1:27017

6 Setup LM Studio (IMPORTANT)

1. Open **LM Studio**
2. Download a model (recommended):
 - `qwen2.5-7b-instruct`
3. Click **Start Server**
4. Ensure:

Port: 1234

7 (Optional) Setup TTS Voices

Place Piper `.onnx` voice files inside:

voices/

Example:

voices/en_US-sam-medium.onnx
voices/en_US-sam-medium.onnx.json

Running the Project (EVERY TIME)

1 Activate Virtual Environment

`.\env\Scripts\Activate.ps1`

2 Start Streamlit App

`streamlit run streamlit_app.py`

3 Open Browser

If it doesn't auto-open, go to:

http://localhost:8501

How to Use the App

1. Click  **Start Camera**
 2. Sign ASL letters clearly
 3. Letters accumulate in the buffer
 4. Click  **Stop Camera**
 5. System will:
 - Refine sentence (LLM)
 - Translate (if enabled)
 - Speak sentence (if TTS enabled)
 - Save data to MongoDB
-

Before Closing VS Code

- Stop Streamlit with `Ctrl + C`
 - Close LM Studio
 - You're safe to exit
-

Quick Start (April Demo Checklist)

- ✓ Open VS Code
 - ✓ Open project folder
 - ✓ Activate venv
 - ✓ Start MongoDB
 - ✓ Open LM Studio → Start server
 - ✓ `streamlit run streamlit_app.py`
 - ✓ Open browser → Demo
-

Notes for Evaluation / Demo

- Fully **local AI system**
- No cloud APIs
- Real-time multimodal processing
- Scalable to word-level and sentence-level ASL
- Strong accessibility & inclusion use-case

Common Issues

LM Studio not running

Start LM Studio and server on port 1234

Camera not opening

Change camera index (0 / 1 / 2)

TTS not playing

Check voice files in [voices/](#)

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Isha's Personal Steps to Run the Project (Next Time)

Step 0: Before You Start (2-minute check)

Make sure:

-  Laptop camera is free (no Zoom / Meet running)
 -  Internet not required (project is local)
 -  LM Studio is installed
 -  MongoDB is installed
-

Step 1: Open Project in VS Code

1. Open **VS Code**

2. Click **File** → **Open Folder**
3. Select:

Multimodal-Realtime-Sign-Language-to-Text-and-Speech-Translator-main

4. Open **Terminal** in VS Code
(**Ctrl + ~**)

Step 2: Activate Virtual Environment

In the VS Code terminal, type **exactly**:

```
.\env\Scripts\Activate.ps1
```

What to check

- You **MUST** see:

```
(venv)
```

before the path

If you don't → STOP and fix this first.

Step 3: Start MongoDB

Option A (Recommended – Automatic)

- MongoDB usually starts automatically on Windows
- No command needed

Option B (If MongoDB not running)

Open **MongoDB Compass** OR run:

```
mongod
```

What to check

No error saying “cannot connect to MongoDB”.

Step 4: Start LM Studio (VERY IMPORTANT)

1. Open **LM Studio**
2. Load your model
(example: `qwen2.5-7b-instruct`)
3. Click **Start Server**

What to check inside LM Studio

- Server running
- Port = **1234**
- Status shows **Running**

 Do NOT skip this step
(The project will fail without this)

Step 5: Quick Environment Test (Optional but Safe)

In the same terminal:

```
python -c "import mediapipe, streamlit, cv2, torch; print('OK')"
```

What to check

You should see:

OK

 TensorFlow warnings are NORMAL — ignore them.

Step 6: Run the Project (Main Command)

Type **exactly**:

```
streamlit run streamlit_app.py
```

Step 7: Open the App

- Browser will open automatically
OR
- Manually open:

`http://localhost:8501`

What to check on the webpage

- Camera controls visible
 - Sidebar visible
 - No red error messages
-

Step 8: How to Use During Demo

1. Click  **Start Camera**
 2. Show ASL letters clearly
 3. Letters appear in **buffer**
 4. Click  **Stop Camera**
 5. System will:
 - Refine sentence (LLM)
 - Translate (if enabled)
 - Speak sentence (TTS)
 - Save to MongoDB
-

Step 9: If You Want Speech Output

In sidebar:

-  Enable **Text-to-Speech**
 - Select a voice
 - Keep speed at **1.0**
-

Step 10: Properly Stop Everything

Stop Streamlit

In terminal:

Ctrl + C

Close LM Studio

Just close the app

Close VS Code

Safe to exit now



Quick Troubleshooting (ONLY if something breaks)

✗ “LM Studio not running”

- ✓ Open LM Studio
 - ✓ Start server on port **1234**
-

✗ Camera not opening

- Change camera index (0 → 1 → 2)
 - Close Zoom / Meet
-

✗ No audio

- Check **voices/** folder
 - Try another voice
 - Make sure TTS is enabled
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30-Second April Demo Checklist

- ✓ Open VS Code
 - ✓ Activate venv
 - ✓ Start MongoDB
 - ✓ Open LM Studio → Start server
 - ✓ streamlit run streamlit_app.py
 - ✓ Browser opens → Demo
-

If you want, next I can:

-  Give **exact lines to say during demo**
-  Create a **1-slide architecture explanation**
-  Prepare **mentor questions & answers**

Just say the word 